DIGITAL IMAGE PROCESSING

LECTURE # 8

IMAGE ENHANCEMENT IN SPATIAL DOMAIN-V

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Topics to Cover

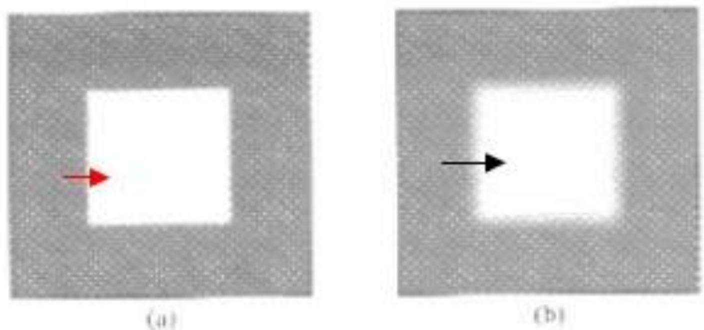
- Gradient Operators
  - **Noise Factor**
  - **Blurred Edges**
  - **Canny Edge Detection**

Noise in an image

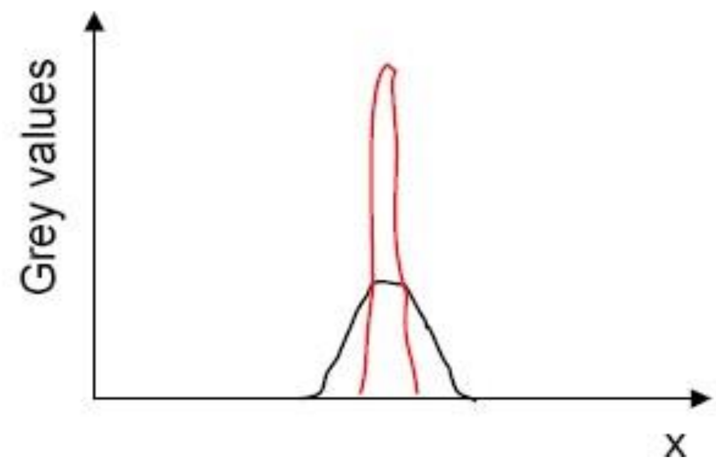
- Problem with Edge Localization if threshold is assume
  - High Threshold may suppress meaningful edges
  - Low Threshold may include unwanted edges
  - Noise may have high magnitude

Blurred Edges

– The detected edge can be rather broad in the case of diffuse edges, resulting in a thick band of pixels instead of a single point of maximum gradient.



Slightly blurred   Heavily blurred



Canny Edge Detection

- The **Canny** edge detection operator was developed by John F. Canny in 1986 and uses a multi-stage algorithm to detect a wide range of edges in images.

- Canny's aim was to discover the optimal edge detection algorithm. In this situation, an "optimal" edge detector means:

  - ✓ *good detection* – the algorithm should mark as many real edges in the image as possible.
  - ✓ *good localization* – edges marked should be as close as possible to the edge in the real image.
  - ✓ *minimal response* – a given edge in the image should only be marked once, and where possible, image noise should not create false edges.

Canny Edge Detection

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- **Algorithm Steps**

  - Image smoothing
  - Gradient computation
  - Edge direction computation
  - Non-maximum suppression
  - Hysteresis Thresholding

Image Smoothing using Gaussian Filter Coefficient

- Reduces image noise that can lead to erroneous output
- Performed by convolution of the input image with a Smoothing filter

$$\frac{1}{159}\begin{bmatrix} 2 & 4 & 5 & 4 & 2 \\ 4 & 9 & 12 & 9 & 4 \\ 5 & 12 & 15 & 12 & 5 \\ 4 & 9 & 12 & 9 & 4 \\ 2 & 4 & 5 & 4 & 2 \end{bmatrix}$$

Image Smoothing

Gradient Computation

- Determines intensity changes
- High intensity changes indicate edges
- Performed by convolution of smoothed image with masks to determine horizontal and vertical derivatives

-1	0	1
-2	0	2
-1	0	1

∇_x

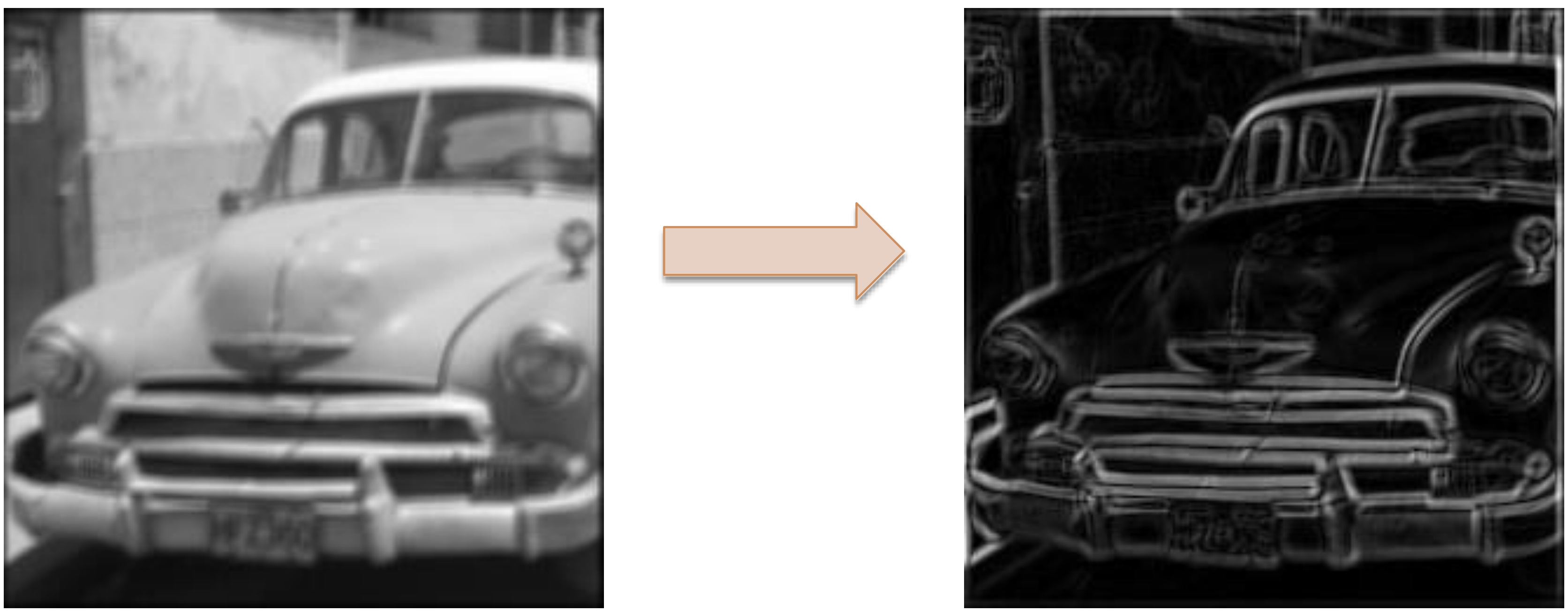
-1	-2	-1
0	0	0
1	2	1

∇_y

Gradient Computation

- Gradient magnitude determined by adding X and Y gradient images

$$|\nabla| = |\nabla_x| + |\nabla_y|$$



Edge Direction Computation

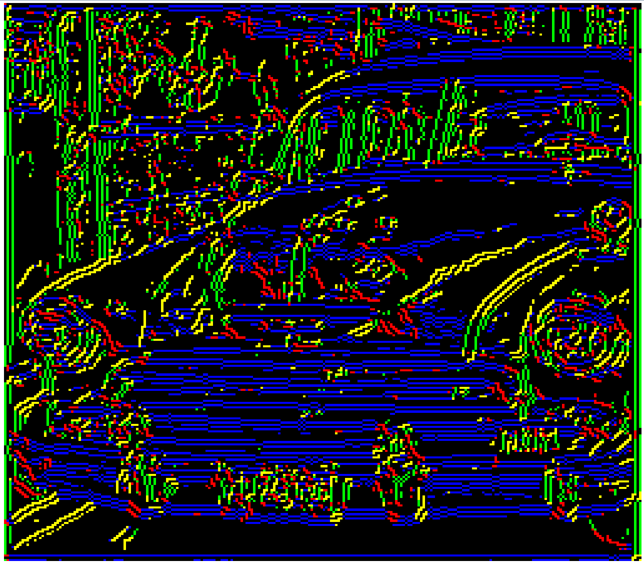
- Edge directions are determined from running a computation on the X and Y gradient images

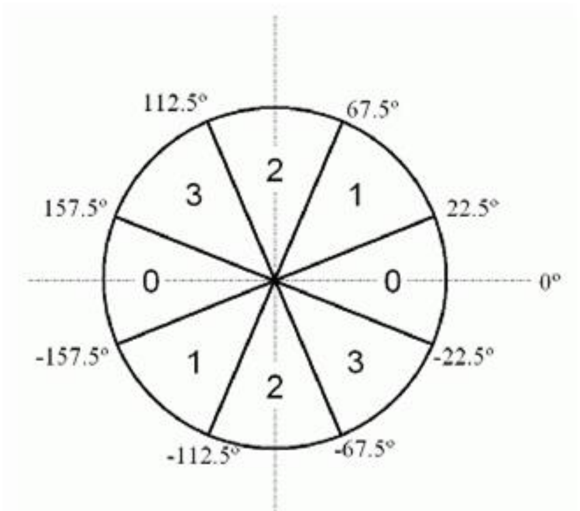
$$\Theta_{x,y} = \tan^{-1}\frac{\nabla_x}{\nabla_y}$$

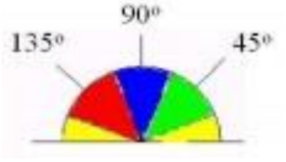
- Edge directions are then classified by their nearest 45° angle

Edge Direction Computation

- Compute θ' by rounding the angle θ to one of four directions 0°, 45°, 90°, or 135°.
- Obviously for edges, $180^\circ = 0^\circ$, $225^\circ = 45^\circ$, etc.
- This means θ in the ranges $[-22.5^\circ$ to $.22.5^\circ]$ and $[157.5^\circ$ to $202.5^\circ]$ would "round" to $\theta' = 0^\circ$.
- For a pictoral representation, each edge take on one of for colors: Here, the colors would repeat on the lower half of the circle (green around 225°, blue around 270°, and red around 315°)







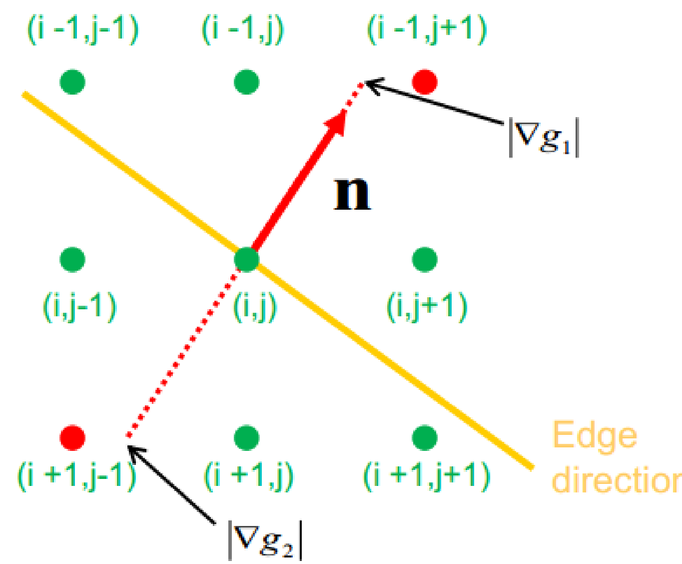
Non-Maximum Suppression

Non-maximum suppression is an edge thinning technique.

- Having found the orientation and magnitude of the gradient for the canny edge detector
- The next step consists of refining the found edges in the image.
- The first refinement is to use non-maximum suppression that thins the found edges to exactly one pixel

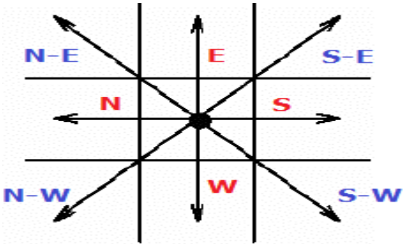
Non-Maximum Suppression

- The green and red dots you see in the image above are pixels.
- We consider a 3×3 neighborhood around the each edge pixel and try to thin it to a 1 pixel wide edge.
- For this, we follow the gradient in both directions at a pixel (i,j) till we are in the next ”row” of pixels (∇g2 and ∇g1).
- Since these points lie between two pixels we have to interpolate their values and compare it to the value at pixel (i,j).
- The pixel at (i,j) is the maximum, if |∇g(i,j)| > |∇g1| and |∇g(i,j)| > |∇g2|.
- The other both pixel are set to zero.



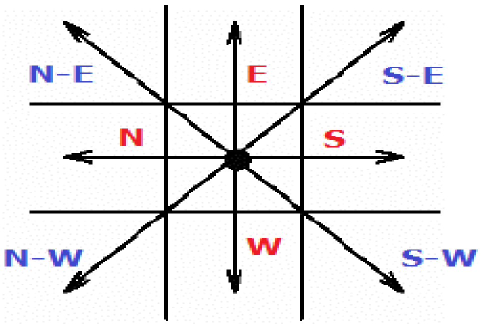
Non-Maximum Suppression

- At every pixel, it suppresses the edge strength of the center pixel (by setting its value to 0) if its magnitude is not greater than the magnitude of the two neighbors in the gradient direction.
  - if the rounded angle is zero degrees (i.e. the edge is in the north-south direction) the point will be considered to be on the edge if its intensity is greater than the intensities in the **west and east** directions,
  - if the rounded angle is 90 degrees (i.e. the edge is in the east-west direction) the point will be considered to be on the edge if its intensity is greater than the intensities in the **north and south** directions,
  - if the rounded angle is 135 degrees (i.e. the edge is in the northeast-southwest direction) the point will be considered to be on the edge if its intensity is greater than the intensities in the **north west and south east** directions,
  - if the rounded angle is 45 degrees (i.e. the edge is in the north west–south east direction) the point will be considered to be on the edge if its intensity is greater than the intensities in the **north east and south west** directions.

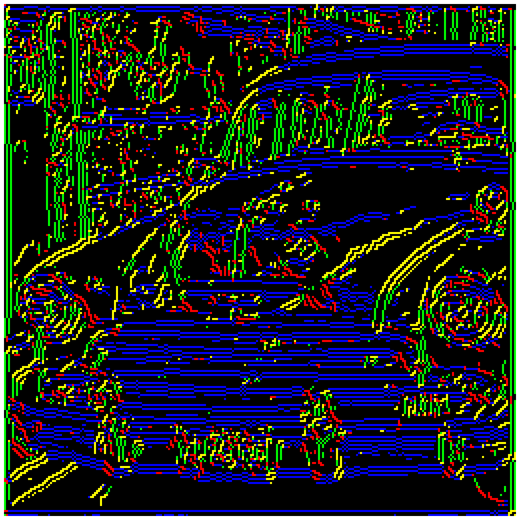


Non-Maximum Suppression

- If pixel (x, y) has the highest gradient magnitude of the three pixels examined, it is kept as an edge.
- If one of the other two pixels has a higher gradient magnitude, then pixel (x, y) is not on the "center" of the edge and should not be classified as an edge pixel.
- The "non-maximal suppression" step keeps only those pixels on an edge with the highest gradient magnitude.
- These maximal magnitudes should occur right at the edge boundary, and the gradient magnitude should fall off with distance from the edge.



Non-Maximum Suppression



Thresholding

- Reduce number of false edges by applying a threshold T
  - all values below T are changed to 0
  - selecting a good values for T is difficult
  - some false edges will remain if T is too low
  - some edges will disappear if T is too high

Thresholds

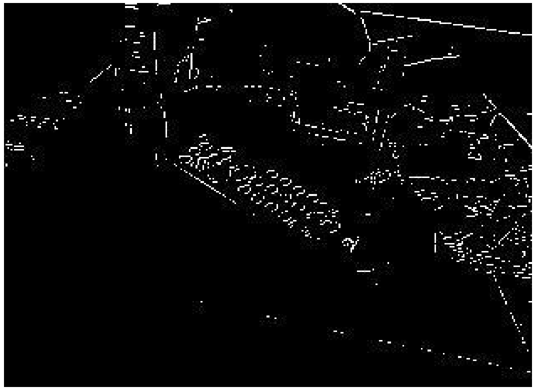
- Thresholds are important, done before or during edge detection.

original image

very high threshold

Thresholds

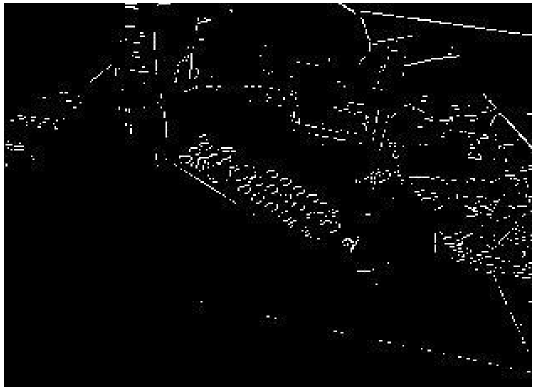
original image



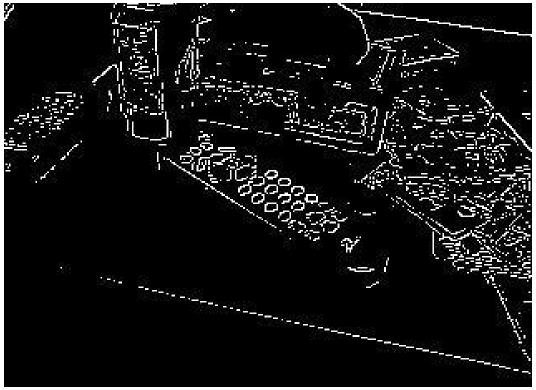
very high threshold

Thresholds

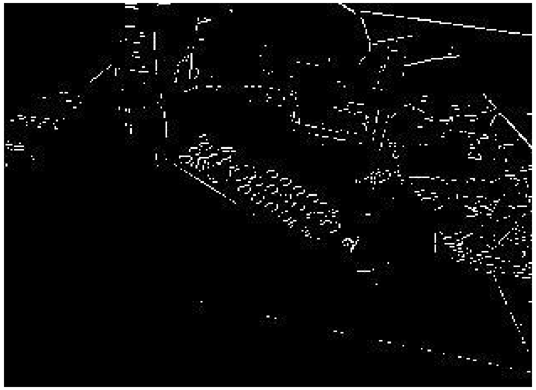
original image



very high threshold

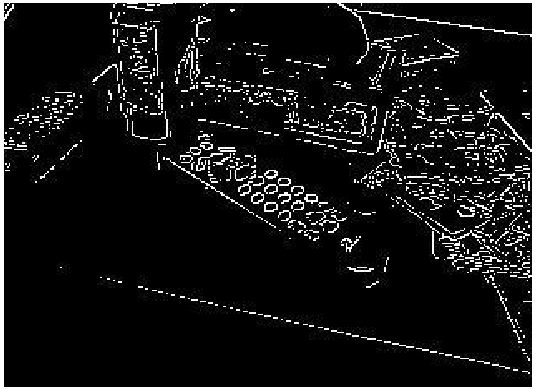


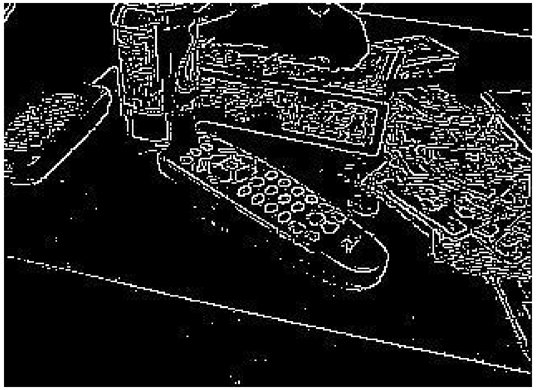
Thresholds



original image

very high threshold





reasonable

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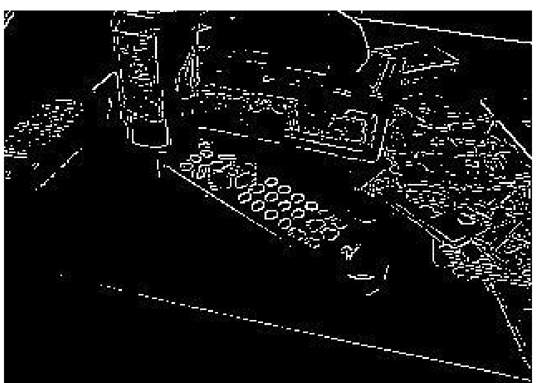
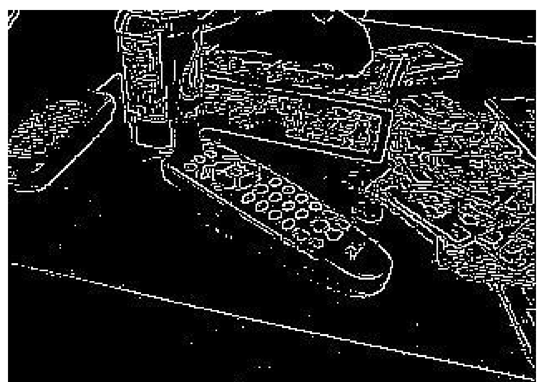
Thresholds

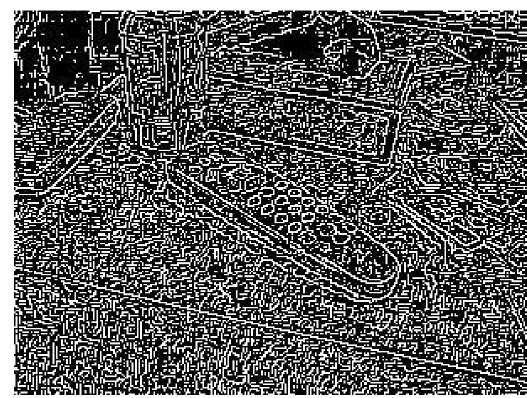
original image

very high threshold →





→ reasonable →



too low !

this all takes time...

Thresholding

- Thresholding with hysteresis requires two thresholds - high and low.
- we begin by applying a high threshold. This marks out the edges we can be fairly sure are genuine. Starting from these, using the directional information derived earlier, edges can be traced through the image.
- While tracing an edge, we apply the lower threshold, allowing us to trace faint sections of edges.
- Once this process is complete we have a binary image where each pixel is marked as either an edge pixel or a non-edge pixel.

Hysteresis Thresholding

- If pixel (x, y) has gradient magnitude less than T_L discard the edge (write out black).

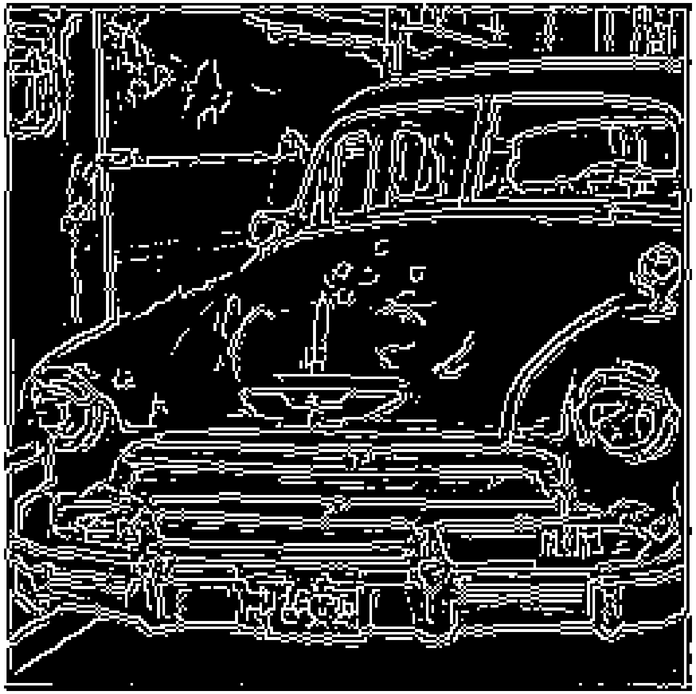
- If pixel (x, y) has gradient magnitude greater than T_H keep the edge (write out white).

- If pixel (x, y) has gradient magnitude between T_L and T_H and any of its neighbors in a 3 × 3 region around it have gradient magnitudes greater than T_H, keep the edge (write out white).

- If none of pixel (x, y)'s neighbors have high gradient magnitudes but at least one falls between T_L and T_H, search the 5 × 5 region to see if any of these pixels have a magnitude greater than T_H. If so, keep the edge(write out white).

- Else, discard the edge (write out black).

Hysterisis Thresholding



References

1. DIP by Gonzalez

For any query Feel Free to ask